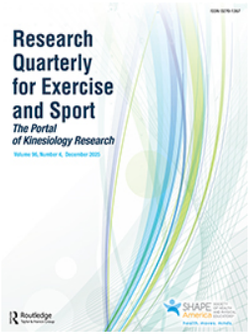

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Beyond Averages: Uncovering Within-Person Links Between Sleep and Performance in Division I Collegiate Football Players

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ABSTRACT

A robust body of research links greater sleep duration and quality to improved athletic performance and competitive outcomes. However, many athletes, particularly collegiate football players, struggle to achieve optimal sleep and accurately assess its quality. Despite the known sleep—performance relationship, little is known about how these variables manifest in real time among student-athletes. This study examined daily associations between objectively measured sleep and athletic performance in National Collegiate Athletic Association Division I football players. Sixty-five athletes aged 17 to 23 years (M = 19.88, SD = 1.41), representing a range of academic years and position groups, wore sensor-based devices over a three-week period to capture sleep metrics (sleep efficiency, latency, and total sleep time) and performance indicators (maximum acceleration, maximum velocity, and explosive movement). Multilevel modeling revealed no significant between-person effects, suggesting that athletes who slept better on average did not necessarily perform better on average. However, within-person analyses indicated that nights with longer sleep latency (estimate, −.007; 95% BCI, −.013, −.003) or lower sleep efficiency (estimate, .005; 95% BCI, .001, .010) predicted reduced maximum acceleration the next day. Conversely, days with lower maximum acceleration predicted shorter sleep latency (estimate, 6.869; 95% BCI, 3.998, 9.269) and higher sleep efficiency (estimate, −5.289; 95% BCI, −10.170, −1.027) that night. These findings underscore a dynamic, bidirectional relationship between sleep and performance at the daily level and highlight the need for individualized, athlete-centered sleep interventions that extend beyond sleep duration to include routine assessment, comprehensive education, and strategies to mitigate sleep disruptors.

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Sleep is a critical component of athletic performance, recovery, and overall health (Watson, 2017). Adequate sleep has been associated with improved reaction time, cognitive function, muscle recovery, and physical performance (Charest & Grandner, 2020; Cunha et al., 2023; Kroshus et al., 2019; Penggalih et al., 2021). Conversely, insufficient or poor-quality sleep has been linked to increased injury risk, impaired motor and cognitive function, and decreased performance (Cullen et al., 2019; Milewski et al., 2014; Walsh et al., 2021). The relationship between sleep and exercise is also bidirectional, with exercise typically improving sleep quality and duration (Dolezal et al., 2017; Halson, 2014; Kredlow et al., 2015). These connections are particularly relevant for student-athletes, who face rigorous training schedules, high physical exertion, and significant psychological stressors that can alter sleep patterns (Mah et al., 2018; Nédélec et al., 2015). Overtraining has also been linked to reduced parasympathetic activity and heightened sympathetic activity during sleep, further diminishing sleep quality (Burton et al., 2010; Pichot et al., 2002).

Despite the well-established importance of sleep, many student-athletes fail to meet recommended guidelines. The American Academy of Sleep Medicine and Sleep Research Society suggest young adults benefit most from at least nine hours of sleep per night (Watson et al., 2015), yet collegiate athletes often average far less due to demanding schedules, late practices, travel, and competition (Fullagar et al., 2015; Gupta et al., 2017; Nédélec et al., 2015; Watson, 2017). Football players in particular experience insufficient sleep and training fatigue, with FBS football athletes reporting the highest prevalence of sleep difficulties among National Collegiate Athletic Association (NCAA) sports, averaging just under six hours per night (National Collegiate Athletic Association, 2019). Professional football players also experience sleep-related health concerns, such as obstructive sleep apnea, which further impairs recovery and functioning (Albuquerque et al., 2010; George et al., 2003). Given the links between insufficient sleep, increased injury risk, and reduced recovery, football players represent a high-risk group in need of more applied research on sleep and performance (Raikes et al., 2019).

Although research has established general associations between sleep and performance, there is a lack of high-quality studies examining effective sleep interventions for athletes, particularly within real-world training environments (Cunha et al., 2023; Walsh et al., 2021). Previous work has often used experimental designs to study acute sleep restriction or exercise manipulations, which provide important insights but do not reflect the day-to-day variability athletes face (Fullagar et al., 2015; Simpson et al., 2017). The ecological momentary assessment (EMA) framework, which involves

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repeated measurements of behaviors and physiological responses in naturalistic contexts, has been underutilized in sport research. Very few studies have examined the bidirectional relationship between sleep and performance in collegiate athletes using objective methods across multiple days of training.

Wearable technology offers a unique opportunity to address these gaps by providing continuous, objective data on both sleep and training. Unlike self-reports, which are subject to recall bias, wearables capture detailed metrics such as sleep onset, duration, efficiency, and exertion during exercise. Research has shown the Oura Ring provides valid measurement across multiple sleep variables when compared to polysomnography, the gold standard for sleep monitoring, although it underestimates sleep efficiency and REM sleep (Svensson et al., 2024). Studies using Oura have highlighted the benefits of wearable technology for obtaining individualized, real-time sleep data that can inform both monitoring and intervention (Bigalke et al., 2025; Kung et al., 2025). Similarly, Catapult microtechnology units have demonstrated strong reliability in measuring acceleration, deceleration, change of direction, and jumps, making them a leading tool for performance monitoring in elite sport (Crang et al., 2022; Mackay et al., 2025; Nicolella et al., 2018). Together, these devices provide complementary measures that capture sleep quality and neuromuscular performance, advancing the accuracy and ecological validity of sleep-performance research in sport.

The current study aimed to expand the understanding of the relationship between sleep and athletic performance by examining daily associations between sleep metrics and training performance in NCAA Division I football players. By leveraging wearable technology (Oura for sleep, Catapult for performance) over multiple weeks of training, this study applies an ecologically valid, daily within-person framework that addresses limitations of self-report and laboratory-based designs. This unique contribution provides novel insight into the bidirectional relationship between sleep and performance and offers a foundation for developing targeted interventions to optimize sleep quality, athletic performance, and well-being in collegiate football programs.

Methods

Participants

The sample included 65 male NCAA Division I football players, aged 17 to 23 years ($M = 19.88$, $SD = 1.41$), representing a range of position groups. Participant weights ranged from 169.4 to 363.2 pounds ($M = 233.15$, $SD = 47.16$). Academic standing was distributed as follows: freshman (13.8%), sophomore (33.8%), junior (18.5%), senior (18.5%), and graduate student (15.4%). Self-identified race/ethnicity included Black (58.5%), White (27.7%), Hispanic (6.2%), and Multiracial (7.7%). Full demographic details are presented in Table 1. Participation was voluntary and there were no exclusion criteria.

Procedure

Variables were chosen and data were collected as part of the regular program operations of an NCAA collegiate football team. This project was determined to be exempt from the University Institutional Review Board because it involved retrospective analysis of de-identified data collected during routine program operations for performance monitoring and improvement. No additional data were collected for research purposes, and all analyses were conducted on de-identified records. The research team retrospectively analyzed data on sleep variables, measured using Oura wearable technology, and performance variables,

Table 1. Demographic, sleep, and exercise characteristics.

Gender, n (%)	
Male	65 (100%)
Female	0 (0%)
Age, Mean (SD), Years	19.88 (1.41)
Race, n (%)	
White	18 (27.69%)
Black	38 (58.46%)
Hispanic	4 (6.15%)
Mixed	5 (7.69%)
Class, n (%)	
Freshman	9 (13.85%)
Sophomore	22 (33.85%)
Junior	12 (18.46%)
Senior	12 (18.46%)
Graduate	10 (15.38%)
Weight, mean (SD), pounds	233.15 (47.16)
Sleep Metrics	
Total Sleep Time, mean (SD), hours	6.02 (2.68)
Sleep Onset Latency, mean (SD), minutes	16.52 (14.64)
Sleep Efficiency, mean (SD), percentage	80.91 (21.23)
Performance Metrics	
Max Velocity, mean (SD), mph	13.16 (6.16)
Max Acceleration, mean (SD), m/s^2	3.96 (0.90)
Explosive Efforts, mean (SD)	22.60 (12.90)

tracked through Catapult technology, over a three-week period (i.e., July 30, 2023 - August 20, 2024). This period (i.e., fall training camp), was chosen intentionally due to the more consistent daily training demands compared to the variability of an in-season schedule. Fall training camp included approximately 2 hours of morning training for four days on, one day off, repeated throughout the three week period. Not all participants contributed the same number of data days for various reasons (e.g., not wearing an Oura ring the evening prior to practice, not being cleared to participate in practice the next day)

Measures

Sleep variables

Sleep was measured using wearable Oura rings (Oura Health, Oulu, Finland), which has shown high accuracy in wake-sleep detection and approximately 79% agreement with polysomnography, the gold standard in sleep research (Altini & Kinnunen, 2021; Roberts et al., 2019). In addition, a recent multi-night validation study of the Oura Ring found high epoch-level reliability, measured using prevalence- and bias-adjusted kappa (PABAK = 0.83–0.84) and an overall inter-device reliability of 94.8% (Svensson et al., 2024). The ring uses physiological signals (i.e., a combination of motion, nocturnal heart rate and heart rate variability, and pulse wave variability amplitude), in combination with sophisticated machine learning-based methods, to compute sleep/wake states (de Zambotti et al., 2019). The three sleep metrics utilized in the current study include: 1) sleep efficiency (i.e., the percentage of time spent asleep relative to time spent in bed); 2) sleep latency (i.e., the duration of time, in minutes, it takes to fall asleep); and 3) total sleep time (i.e., the total number of hours spent asleep during the night).

Performance variables

Training performance was measured using Catapult athlete monitoring system technology, a wearable device that tracks a range of athletic movements. Catapult units are compact, wearable devices equipped with a GPS receiver, accelerometer, gyroscope, and magnetometer, which collectively provide detailed data on the volume, rate, and nature of athletic movement. Catapult devices have demonstrated excellent intra-device reliability (coefficient of variation [CVs] = 0.01–3.9%; intraclass correlation coefficient [ICCs] = 0.77–1.00), though reliability is more variable for threshold-based acceleration and deceleration counts (Crang et al., 2022; Nicolella et al., 2018). These devices are widely used by collegiate and professional sports teams to inform practice planning, monitor athlete load, build athlete profiles, and aid in the return-to-play process post-injury (Akyildiz et al., 2022; Boyd et al., 2011; Luteberget et al., 2018; Meylan et al., 2017). The three performance metrics utilized in the current study include: 1) maximum acceleration—the highest recorded rate of acceleration achieved during the activity, measured in m/s^2; 2) maximum velocity—the highest recorded speed in miles per hour; and 3) explosive movements—the total number of high-intensity accelerations, decelerations, and changes of direction during the activity.

Data analyses

All analyses were conducted using Mplus version 8.11 (Muthén & Muthén, 1998–2017). We used multilevel Bayesian models because data consisted of multiple daily measurements nested within each athlete. These models allowed us to examine: 1) within-person effects—how day-to-day changes in an individual's sleep relate to changes in their exercise performance, and vice versa; and 2) between-person effects—how differences in average sleep between individuals relate to differences in average exercise performance.

Bayesian estimation combines data observed with prior beliefs (or assumptions) about likely values of the parameters. This approach gives a probability distribution for each effect, rather than a single point estimate. From this, the median estimate, a 95% Bayesian credible interval (BCI; the range where the effect is likely to fall with 95% probability), and the proportion of the distribution above or below zero can be identified. Unlike traditional statistics, Bayesian methods do not rely on p-values, but the credible intervals provide an intuitive way to determine whether an effect is likely to exist. 50,000 iterations, two chains, and a thinning interval of 10 were utilized. This ensured that the model had enough samples to accurately estimate the effects and reduced the chance that consecutive samples were too similar. Convergence was checked by confirming that the potential scale reduction factor was near 1 and by visually inspecting trace plots to make sure the estimates were stable.

Missing observations occurred when participants did not wear the Oura ring or were unavailable for practice. These missing data were handled using Bayesian estimation in Mplus, which includes all available data in the analysis and provides unbiased parameter estimates under the assumption that data are missing at random. Separate models were run for each combination of sleep and exercise variables, resulting in 18 models. To address concerns about the potential for false positives due to multiple testing, 99% BCIs in addition to the standard 95% BCIs were reported for all significant effects. Effects that remained significant at the 99% BCI were considered more robust, providing greater confidence that these findings are not the result of chance.

Results

Descriptive statistics

Demographic information, as well as means and standard deviations of all sleep and exercise variables, are presented in Table 1.

Models testing bidirectional effects of sleep and exercise

Between-person effects

Model convergence was achieved for all models with final potential scale reduction values ≤1.002. The intraclass correlation coefficients, when averaged across the three models for

Table 2. Sleep predicting next-day exercise.

	Max velocity			Max acceleration			Explosive efforts		
Predictor	Est.	SD	95% BCI	Est.	SD	95% BCI	Est.	SD	95% BCI
Total Sleep Time									
Within	−.101	.221	−.424 to .366	.031	.020	−.007 to .070	.119	.273	−.503 to .658
Between	−.114	.400	−.862 to .805	−.178	1.555	−4.661 to 2.473	−1.642	1.822	−5.468 to 2.266
Sleep Onset Latency									
Within	−.003	.033	−.069 to .063	**−.007**	**.003**	**-.013 to −.003**	.024	.047	−.070 to .116
Between	.174	.150	−.177 to .438	.090	.076	−.003 to .301	−.321	.785	−1.857 to 1.435
Sleep Efficiency									
Within	.010	.026	−.030 to .066	**.005**	**.002**	**.001 to .010**	−.010	.033	−.084 to .060
Between	.019	.312	−.488 to .784	−.022	2.094	−5.509 to 4.151	−.464	3.722	−9.459 to 8.905

Table 3. Exercise predicting same-night sleep.

	Total sleep time			Sleep onset latency			Sleep efficiency		
Predictor	Est.	SD	95% BCI	Est.	SD	95% BCI	Est.	SD	95% BCI
Max Velocity									
Within	.030	.037	−.055 to .086	.073	.193	−.349 to .510	−.032	.320	−.735 to .526
Between	−.026	.139	−.376 to .212	.922	.838	−.284 to 3.291	−.099	.887	−2.110 to 1.684
Max Acceleration									
Within	−.278	.315	−.940 to .309	**6.869**	**1.320**	**3.998 to 9.269**	**−5.289**	**2.481**	**-10.170 to −1.027**
Between	−.233	.351	−.959 to .468	3.100	1.500	−.972 to 5.436	−.944	2.216	−5.499 to 3.010
Explosive Efforts									
Within	.017	.017	−.018 to .049	−.167	.095	−.317 to .045	.138	.146	−.119 to .464
Between	.018	.029	−.046 to .084	−.040	.155	−.326 to .292	.175	.178	−.226 to .471

each outcome variable, indicated that 12.07% of the variance in total sleep time, 6.42% in sleep onset latency, 5.39% in sleep efficiency, 5.27% in max velocity, 47.07% in max acceleration, and 48.30% in explosive efforts existed between participants. There were no significant between-person associations between sleep and exercise (see Tables 2 and 3). That is, after accounting for individual differences, total sleep time, sleep onset latency, and sleep efficiency did not significantly predict average levels of max velocity, max acceleration, and explosive efforts across participants, and max velocity, max acceleration, and explosive efforts did not significantly predict average levels of total sleep time, sleep onset latency, and sleep efficiency across participants. Thus, inter-individual differences in sleep did not predict exercise and inter-individual differences in exercise did not predict sleep.

Within-person effects

The intraclass correlation coefficients, when averaged across the three models for each outcome variable, indicated that 87.93% of the variance in total sleep time, 93.58% in sleep onset latency, 94.61% in sleep efficiency, 94.73% in max velocity, 52.93% in max acceleration, and 51.70% in explosive efforts existed across days within participants. There were two significant, within-person effects of sleep on exercise (see Table 2). First, days on which participants had a longer sleep onset latency than they do on average predicted lower next-day max acceleration (estimate, −.007; 95% BCI, −.013, −.003; 99% BCI, −.016, −.002). Specifically, each additional minute of sleep onset latency relative to an athlete's average corresponded to a 0.18% reduction in next-day maximum acceleration. Second, days on which participants had a lower sleep efficiency than they do on average predicted lower next-day max acceleration (estimate, .005; 95% BCI, .001, .010; 99% BCI, .001, .011). Specifically, each percentage point decrease in sleep efficiency relative to an athlete's average corresponded to a 0.13% reduction in next-day maximum acceleration.

There were also two significant, within-person effects of exercise on sleep (see Table 3). First, days on which participants had a lower max acceleration than they do on average predicted shorter sleep onset latency that night (estimate, 6.869; 95% BCI, 3.998, 9.269; 99% BCI, 3.247, 10.717). Specifically, each 1% decrease in maximum acceleration relative to an athlete's average corresponded to a decrease of 0.27 minutes in sleep onset latency that night. Second, days on which participants had a lower max acceleration than they do on average predicted higher sleep efficiency that night (estimate, −5.289; 95% BCI, −10.170, −1.027; 99% BCI, −10.682, .146). Specifically, each 1% decrease in maximum acceleration relative to an athlete's average corresponded to a decrease of 0.21% points in sleep efficiency that night. Of note, upon examination of the wider 99% BCI as a form of sensitivity analysis given multiple testing, the first three within-person effects remained significant, but the effect of lower maximum acceleration predicting higher sleep efficiency that night was no longer significant, suggesting that this finding may be less robust than the others.

Discussion

While the link between sleep and athletic performance is well-documented (Kroshus et al., 2019; Mah et al., 2011; Penggalih et al., 2021), this study extended previous literature by examining daily reciprocal within-person associations between sleep and performance among NCAA Division I football players using wearable technology. Results indicated that

sleep predicted next-day athletic performance, such that individuals had decreased maximum acceleration on days in which their previous-night sleep latency was higher or sleep efficiency was lower than their own average. In addition, athletic performance impacted sleep, such that days in which participants had lower maximum acceleration than their own average predicted lower sleep latency and higher sleep efficiency that night. Notably, no within-person effects emerged for total sleep time nor any between-person effects for any of the sleep and athletic performance variables. The absence of between-person effects suggests that athlete-level averages may mask meaningful day-to-day fluctuations, and that traditional recommendations focused solely on group norms may be insufficient. Instead, individualized monitoring of nightly sleep quality and latency may be more informative for guiding performance interventions.

The present study found that on nights when athletes experienced longer sleep latency or lower sleep efficiency than their own average, they had reduced maximum acceleration at training the following day. The current sample's mean sleep latency (17 minutes) fell within the range of normative adult sleep latency (Littner et al., 2005; Thomas & Anderson, 2013). Conversely, the mean sleep efficiency (81%), fell below the commonly cited 85% benchmark for intact sleep continuity, reflecting broader difficulties athletes face in meeting quality sleep standard (Buysse, 2014; Ohayon et al., 2017; Roberts et al., 2019). Importantly, results indicated that the vast majority of variance in both sleep latency and efficiency occurred across nights within individuals rather than between individuals. This high degree of within-person variability supports previous work showing that college athletes experience fluctuating sleep patterns due to shifting training, academic, and social demands (Gupta et al., 2017; Mah et al., 2018; Watson, 2017). These findings underscore the importance of personalized, real-time feedback and interventions, as population-level averages may overlook the performance consequences of nightly variability.

Previous research has documented associations between insomnia symptoms and impaired cognitive and psychomotor functioning, including slowed reaction times and elevated fatigue (Fortier-Brochu et al., 2012; Riedel & Lichstein, 2000). The current findings uniquely demonstrate that even transient disruptions in sleep latency and efficiency can impair neuromuscular performance the next day. Because explosive athletic movements like maximum acceleration rely on rapid, high-effort motor outputs, sleep disruptions may reduce an athlete's ability to consistently perform at peak intensity. These results also build on evidence linking sleep efficiency to central nervous system recovery and physical readiness (Bonnar et al., 2018; Lastella et al., 2015) and offer new insights into how real-time changes in sleep influence performance in student-athletes.

Our study also revealed that lower maximum acceleration at practice was followed by faster sleep onset and higher sleep efficiency that night. These findings expand upon previous work on the beneficial effects of exercise on sleep initiation and maintenance by showing that daily variations in the quality or intensity of exercise also impacts sleep (Dolezal et al., 2017; Kredlow et al., 2015). A possible explanation for this relationship is that days with lower acceleration reflect fatigue-related performance decrements which may facilitate faster sleep initiation and more efficient sleep. Alternatively, lower physical strain may reduce pre-sleep arousal or muscle tension, contributing to better sleep quality. Interestingly, these results may suggest a form of compensatory sleep response or recovery mechanism on nights following less intense exertion. This interpretation is consistent with the sleep homeostasis model, which posits that sleep pressure accumulates based on prior wakefulness and exertion (Borbély, 1982). However, this hypothesis cannot be confirmed with the current data and future studies could investigate whether subjective fatigue or perceived exertion mediates this relationship.

In contrast to previous work highlighting the importance of sleep duration for performance (Bonnar et al., 2018; Mah et al., 2011), this study did not observe significant associations between total sleep time and any performance outcomes. This discrepancy may be due to the unique physiological demands of football. Explosive movements and peak acceleration/velocity markers may rely more on neuromuscular efficiency than aerobic endurance (Turner & Stewart, 2014). As such, sleep quality may play a larger role than sleep duration in facilitating the kind of motor coordination and reaction speed necessary for these particular metrics. Alternatively, this divergence could reflect methodological differences given that much of the existing literature has relied on self-reported sleep and/or average scores across time, while the current study captured objective data focusing on deviations from individuals' own average scores. Nonetheless, these are solely speculations that cannot be confirmed by the existing data, and future research should continue to examine whether sleep quality or duration is more strongly linked to performance measures and why.

Clinical implications

This study offers several clinical implications for sports medicine professionals, coaches, and athletic administrators aiming to optimize athlete performance and well-being. Chronic overexertion is common among collegiate athletes (Vetter & Symonds, 2010), particularly Division I football players, and this study highlights the need for athletic departments to prioritize overall sleep health, not just quantity but also quality. Our findings suggest that improving sleep latency and efficiency, rather than solely extending sleep duration, effectively supports day-to-day neuromuscular performance (Mah et al., 2011; Penggalih et al., 2021). Integrating sleep assessments into preparticipation exams and conducting regular time demands surveys can improve understanding and address sleep-related risks (Kroshus et al., 2019). Specific practices include reviewing athlete sleep data on a regular basis and establishing thresholds that flag concerning deviations from individual baselines to inform training adjustments. While wearable technologies offer a practical, individualized method for monitoring within-person fluctuations in sleep and guiding recovery strategies (de Zambotti et al., 2019; Fullagar et al., 2015), their feasibility and costs must be considered when applied at scale across teams and programs. Improving sleep quality requires a multifaceted approach that includes athlete education (e.g.,

impact of substance use on sleep), behavioral tools (e.g., sleep masks, blue light glasses), and evidence-based interventions such as Cognitive Behavioral Therapy for Insomnia and mindfulness-based treatments (Cunha et al., 2023; Silvani et al., 2022; Taylor & Pruiksma, 2014; Wang et al., 2020). Embedding sleep health components into training and athlete wellness programs may yield long-term benefits for both health and performance.

Strengths and limitations

The current study had several strengths, including a well-representative sample of Division I collegiate football players in terms of age, race, academic year, weight, and position group. The study incorporated daily measurements of sleep and performance across three weeks of training, providing a more granular view of athlete functioning. Diverse metrics were incorporated which allowed for a comprehensive examination of how specific dimensions of sleep relate to distinct performance outcomes. Lastly, the use of wearable technology (i.e., Oura rings for sleep, Catapult for performance) offered objective, high-resolution data, improving upon the limitations of traditional self-report measures.

The current study also had several limitations. Generalizability is restricted given that data were collected from a single team, sport, and gender. Data may also be influenced by performance bias, as participants were aware their sleep and performance data were being monitored. Although the assessment period was strategically chosen during a time of relatively consistent training demands, daily performance remains opportunity-driven, and a three-week window restricts the ability to assess long-term effects of sleep on athletic performance. In addition, because data were collected as part of regular program operations, rather than a prospective study designed by the research team, causal interpretations are limited and the retrospective nature should be acknowledged. Further, unmeasured external factors such as illness or injury, nutrition, stimulant use, environmental noise, and psychological stress may have influenced both sleep and performance outcomes.

Although wearable technology enables ecologically valid, continuous monitoring, we observed high variability in some measures. This variability likely reflects both genuine fluctuations in sleep during training camp and potential measurement error inherent to wearable devices. Validation studies indicate that the Oura Ring shows acceptable agreement with polysomnography and that Catapult devices demonstrate strong intra-device reliability (CVs typically < 4%; ICCs 0.77–1.00; Crang et al., 2022; Nicolella et al., 2018; Svensson et al., 2024). Still, variability should be interpreted cautiously, and replication with multiple measurement approaches is warranted. Future research should expand to include diverse athletic populations, a broader range of sleep and performance variables, extended monitoring periods, and targeted interventions examining how specific sleep dimensions (e.g., efficiency, latency) influence sport-specific outcomes.

Conclusion

The current findings suggest that the relationship between sleep and athletic performance is more complex than previously understood, with evidence highlighting a nuanced, reciprocal within-person link between sleep quality indicators and maximum acceleration. These results point to the need for future research that examines whether such patterns extend across different sports, athlete populations, and performance domains, using larger samples and longer monitoring periods. Integrating wearable data with subjective ratings of fatigue, recovery, and stress may further clarify which sleep parameters most strongly influence sport-specific outcomes. Rather than relying solely on broad sleep extension strategies, targeted interventions to improve sleep efficiency and reduce sleep latency may better support neuromuscular performance. Promoting high-quality sleep remains a worthwhile investment, and programming directed toward both athletes and athletic staff is well warranted.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Author contribution

Tess M. Kilwein played a lead role in conceptualization and data curation and an equal role in visualization, writing—original draft, and writing—review and editing. Kaelene A. Curry played a lead role in investigation, methodology, and project administration; a supporting role in conceptualization; and an equal role writing-original and writing—review and editing. Julie Sutcliffe played a supporting role in conceptualization and an equal role in writing-original draft and writing-review and editing. Charles A. Manzer played a lead role in formal analysis an equal role in visualization, writing-original draft, and writing-reviewing and editing.

Data & code availability statement

All data, analysis code, and research materials are available upon reasonable request.

Statement of ethics/IRB approval

This project was determined to be exempt from the University Institutional Review Board because it involved retrospective analysis of de-identified data collected during routine program operations for performance monitoring and improvement. No additional data were collected for research purposes, and all analyses were conducted on de-identified records

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